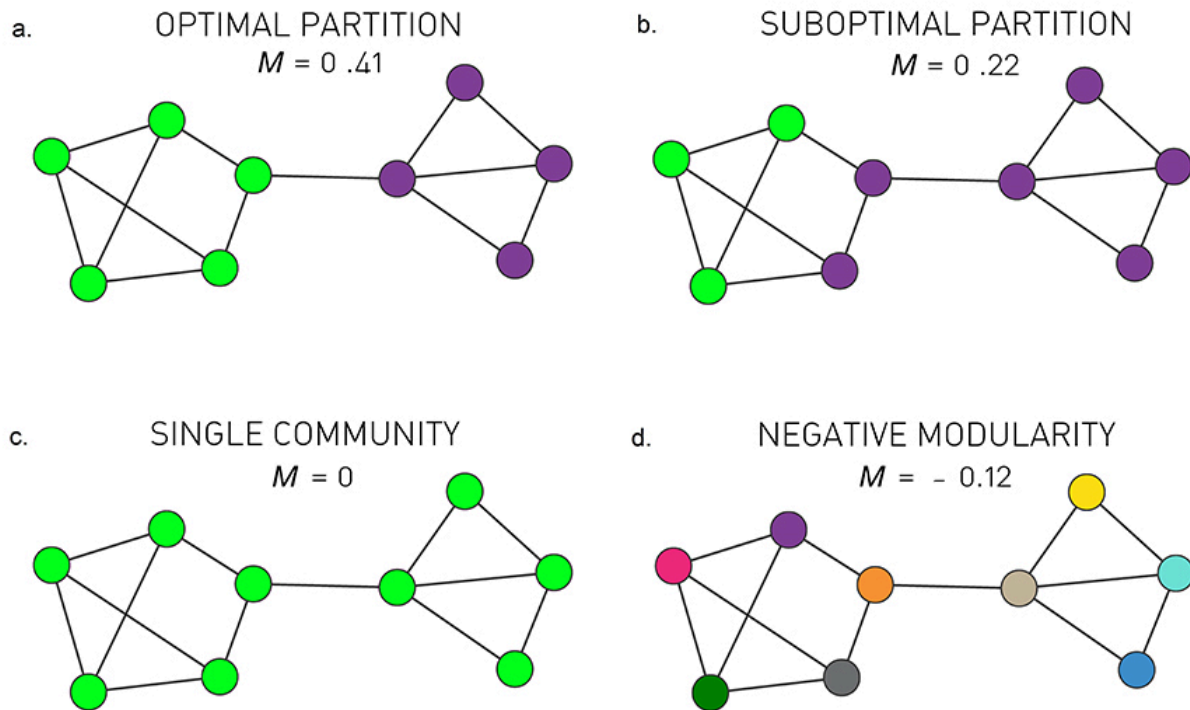


# L7: Modularity Metric – Selecting The Community Assignment



Four partitions of a network: a) optimal partition b) suboptimal partition c) single community d) negative modularity (Image 9.16) from [networksciencebook.com](http://networksciencebook.com)

To get a better intuition about the modularity metric, consider the 9-node network shown in the visualization. Visually, we would expect this network to have two communities: the group of five nodes at the left and the group of four nodes at the right.

The visualization shows four possible community assignments.

The first two assignments have two communities. Assignment (a) is what we would expect visually and it has a modularity of  $M=0.41$ . It turns out that this is the highest possible modularity value for this network.

Assignment (b) is clearly a suboptimal community structure -- and indeed it has a lower modularity value ( $M=0.22$ )

Assignment (c) assigns all nodes to the same community -- resulting in a modularity value of 0. Clearly this is not a statistically significant community.

Finally, assignment (d) assigns each node to its own community, resulting in a negative modularity value.

Now that we have a way to compare different community assignments, we can return to the hierarchical clustering approaches we saw earlier (*both divisive and agglomerative*) and add an extra step, after the construction of the dendrogram:

Each branching point of the dendrogram corresponds to a different community assignment, i.e., a different partition of the nodes in a set of communities. So, we can calculate the modularity at each branching point, and select the branching point that leads to the highest modularity value.

## Modularity Metric – For Directed and/or Weighted Networks

The modularity definition can be easily modified for directed and/or weighted networks.

Consider directed and unweighted networks first. Suppose that the out-degree of node  $i$  is  $k_{i,out}$ , and the in-degree of node  $j$  is  $k_{j,in}$ .

Further, suppose that  $A_{i,j} = 1$  if there is an edge from node  $i$  to node  $j$  – and 0 otherwise.

We can rewrite the modularity definition as:

$$M = \frac{1}{L} \sum_{c=1}^C \sum_{(i,j) \in C_c} (A_{i,j} - \frac{k_{i,out} k_{j,in}}{L})$$

Similarly, if the network is both weighted and directed, suppose that the “out-strength” (i.e., the sum of all outgoing edge weights) of node  $i$  is  $s_{i,out}$ , while the in-strength of node  $j$  is  $s_{j,in}$ .

Further, suppose that  $A_{i,j}$  is the weight of the edge from node  $i$  to node  $j$  – and 0 if there is no such edge.

$$M = \frac{1}{S} \sum_{c=1}^C \sum_{(i,j) \in C_c} (A_{i,j} - \frac{s_{i,out} s_{j,in}}{S})$$

where  $S$  is the sum of all edge weights.

### Food For Thought

Explain why the modularity formula for directed networks has the term  $L$  instead of  $2L$ .